Ho-Jun Lee

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EDUCATION

Hanyang University **Seoul, Korea**
M.S. in Artificial Intelligence Semiconductor (Advisor: Prof. Haejun Chung) *Mar. 2026 – Present*

Kyung Hee University (KHU) **Yongin, Korea**
B.S. in Applied Physics, College of Applied Science *Mar. 2020 – Feb. 2026*

- Summa Cum Laude (Ranked 1st); Ranked 2nd in the College of Applied Science
- GPA: 4.05 / 4.3 (4.29 / 4.5); Major GPA: 4.14 / 4.3 (4.37 / 4.5)

RESEARCH INTERESTS

ML inverse design for optical structures; optimization theory and ML optimization of optical structures; device modeling with optical simulations; AI-based photonics, nanophotonics, quantum optics, optoelectronics, and condensed matter physics

RESEARCH EXPERIENCE

Department of Artificial Intelligence Semiconductor, Hanyang University **Seoul, Korea**
Graduate Researcher (Advisor: Prof. Haejun Chung) *Mar. 2026 – Present*

- Co-first author of a review on metasurface design scales; performed full-wave FDTD benchmarks comparing unit-cell, supercell, and full-metasurface inverse design of a metalens.
- Developing adjoint-based inverse design of dual-band metalenses and ultra-compact on-chip wavelength demultiplexers with Meep on multi-node MPI and GPU clusters.

Metaphotonics Research Lab, KAIST **Daejeon, Korea**
Research Intern, Dept. of Electronic Engineering (Advisor: Prof. Hamza Kurt) *Jan. 2026 – Feb. 2026*

- Studied metasurfaces and silicon photonics.

Nanophotonics Lab, Kyung Hee University **Yongin, Korea**
Research Student, Dept. of Applied Physics (Advisor: Prof. Sun-Kyung Kim) *Aug. 2024 – Dec. 2025*

- Studied metasurfaces and ML optimization.
- Developed a machine learning algorithm for the optimal design of metasurface unit cells by integrating CST and MATLAB.
- Implemented Bayesian optimization to automatically generate the optimal unit-cell structure for a desired wavelength and metal fill fraction.

Next-Generation Optoelectronics Lab, POSTECH **Pohang, Korea**
Research Intern, Dept. of MSE (Advisor: Prof. Changsoon Cho) *Jul. 2024 – Aug. 2024*

- Participated in the fabrication of perovskite-based devices, including solar cells and LEDs.
- Conducted simulations and performance analysis of perovskite devices using the transfer matrix formalism (TMF).
- Learned and applied computational methods to calculate optical parameters of perovskite-based devices.

PUBLICATIONS

1. Eunbi Jang[†], Jiwan Hong[†], **Ho-Jun Lee**[†], Seokju Choi[†], Hyewon Suk, Chanik Kang, Sunae So[*], and Haejun Chung[*], "Scaling Metasurface Design: From Reusable Unit Cells to Full-Metasurface Inverse Design," *Opto-Electronic Science* (2026, accepted; invited paper). [†]Equal contribution.
2. Eun-Joo Lee, Ji-Ho Yi, **Ho-Jun Lee**, In-Sung Jeon, and Sun-Kyung Kim[*], "Microwave Transparent Metamaterials and Their Applications," *Korean Journal of Optics and Photonics* (2025).
3. Eun-Joo Lee, Jun-Young Kim, **Ho-Jun Lee**, Ji-Ho Yi, In-Sung Jeon, Run Hu, Kyoung-Ho Kim[*], and Sun-Kyung Kim[*], "Ultralow-Metal-Content Epsilon-Mu-Near-Zero Metamaterials for Optically Transparent EMI Shielding," *Advanced Optical Materials* (2026).
4. **Ho-Jun Lee**, Eun-Joo Lee, Ji-Ho Yi, and Sun-Kyung Kim[*], "Inverse Design of Microwave-Transparent Metallic Metamaterials via Bayesian Optimization" (in preparation).

CONFERENCE PRESENTATIONS

1. **Ho-Jun Lee**, Eun-Joo Lee, Ji-Ho Yi, and Sun-Kyung Kim*, "Inverse Design of Microwave-Transparent Metallic Metamaterials via Bayesian Optimization," *The 37th OSK Annual Meeting & 2026 Winter Conference*, BEXCO, Busan, Korea, Feb. 2026 (Poster).

HONORS AND AWARDS

Outstanding Poster Award, Undergraduate Thesis Symposium, Kyung Hee University *Dec. 2025*
Academic Excellence Scholarship (Full Tuition, Valedictorian), Kyung Hee University *Aug. 2025*
Second Prize, Entrepreneurship Award, San Jose State University, USA *Jan. 2025*
National Science & Engineering Scholarship (Full Tuition), KOSAF *2024 – 2025*
Academic Excellence Award, Fall 2023, Kyung Hee University *Mar. 2024*
Academic Excellence Scholarship (Full Tuition, Valedictorian), Kyung Hee University *Dec. 2023*
Academic Excellence Scholarship (Partial Tuition Waiver), Kyung Hee University *Dec. 2023*

PROJECTS AND ACTIVITIES

Silicon Valley Innovation & Entrepreneurship Program, San Jose State University, USA *Jan. 2025*
Founder and President, Semiconductor Academic Club 'VANZ', Kyung Hee University *Jan. 2025*
Cadence Full-Custom IC Design & Term Project, Cadence Foundation, Korea *Jul. 2024 – Aug. 2024*
Samsung Shining Star 2024, Samsung Electronics, Korea *Jan. 2024*
AFM Training Internship Program, V-Memory Co., Ltd., Korea *Aug. 2023*
Sputter Vacuum Deposition Training Internship Program, V-Memory Co., Ltd., Korea *Aug. 2023*

SKILLS

Simulation: Meep FDTD (adjoint optimization), Ansys Lumerical FDTD, CST (FEM), COMSOL Multi-physics
Programming: Python (NumPy, MPI), MATLAB **Computing**: multi-node MPI and GPU clusters
Equipment: AFM, sputtering **Languages**: Korean (native), English (TOEFL 5.0 / 6.0)

MILITARY SERVICE

Sergeant, Republic of Korea Marine Corps (honorably discharged) *Aug. 2020 – Feb. 2022*